

SHALINSKOYE, Russian Federation

WMO: 295780

Lat: 55.7173N

Lon: 93.7461E

Elev: 1310

StdP: 14.01

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.0	-27.1	-37.6	0.7	-33.0	-31.7	1.0	-27.1	26.6	17.6	21.4	18.7	2.5	240	0.629

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	83.1	63.9	80.1	63.2	76.8	61.5	67.3	77.5	65.6	74.9	64.0	73.0	6.5	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.8	93.1	71.4	62.1	87.5	69.9	60.5	82.7	68.3	32.6	77.2	31.3	74.7	30.0	73.0	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	18.0	15.6	-37.9	88.8	7.4	2.6	-43.2	90.7	-47.5	92.3	-51.6	93.7	-57.0	95.7		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.4	3.2	8.0	24.8	37.8	46.7	60.8	63.9	60.0	46.7	33.9	15.6	9.7		
	DBStd	23.67	16.45	14.59	11.50	8.41	8.89	7.34	4.56	6.11	7.74	10.00	14.27	15.56		
	HDD50	6907	1450	1175	782	376	173	10	0	4	153	503	1032	1249		
	HDD65	11266	1915	1595	1247	816	568	162	74	178	550	965	1482	1714		
	CDD50	1214	0	0	0	10	72	332	429	313	54	4	0	0		
	CDD65	96	0	0	0	0	2	35	38	21	0	0	0	0		
	CDH74	1298	0	0	0	3	56	599	387	235	18	0	0	0		
	CDH80	252	0	0	0	0	8	131	62	51	0	0	0	0		
(14) Wind	WSAvg	6.5	6.6	6.4	7.1	7.9	7.7	5.5	4.9	4.9	5.6	6.8	7.2	7.3		
(15) Precipitation	PrecAvg	20.1	0.6	0.6	0.7	1.3	1.9	2.6	2.9	3.5	2.1	1.7	1.3	1.1		
	PrecMax	25.4	1.1	1.2	1.9	2.3	3.7	4.3	5.7	6.9	4.7	3.1	2.3	2.0		
	PrecMin	15.7	0.2	0.2	0.3	0.4	0.7	0.6	0.4	1.9	0.3	0.5	0.7	0.2		
	PrecStd	2.4	0.2	0.3	0.4	0.6	0.8	0.8	1.2	1.0	0.9	0.6	0.5	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.4	39.4	54.2	70.6	80.4	87.4	86.5	85.3	77.1	66.1	49.4	37.0		
		MCWB	30.4	31.7	40.7	48.2	56.6	65.1	67.6	66.4	58.4	49.2	42.6	31.3		
	2%	DB	31.6	35.3	47.4	62.6	74.9	83.6	81.8	79.9	70.7	57.1	40.9	34.2		
		MCWB	27.4	30.1	37.8	45.9	55.2	63.7	64.5	65.1	57.4	46.1	35.0	29.7		
	5%	DB	28.3	31.5	43.3	57.1	69.6	80.8	78.7	75.5	65.5	52.8	37.4	31.3		
		MCWB	24.2	26.5	35.3	44.3	52.2	62.5	64.3	63.2	55.2	43.9	32.0	27.6		
	10%	DB	25.0	28.1	40.0	53.3	64.6	77.2	75.5	71.7	60.6	48.2	33.6	28.2		
		MCWB	21.2	24.2	33.1	41.6	50.2	61.0	62.8	61.1	52.0	41.6	30.0	25.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.8	33.2	41.4	51.0	58.5	68.5	69.2	69.3	59.8	50.7	42.8	32.3		
		MCDB	34.0	37.9	50.7	64.2	76.7	82.3	80.5	79.8	72.2	61.0	50.8	35.4		
	2%	WB	27.2	29.7	38.4	47.9	56.4	66.4	67.4	66.7	58.1	47.7	35.9	30.1		
		MCDB	31.1	34.3	47.1	59.2	70.2	78.3	77.3	75.1	68.0	56.0	39.3	33.6		
	5%	WB	24.4	27.1	35.8	45.4	54.1	64.6	65.8	64.7	56.2	44.9	33.2	28.0		
		MCDB	28.6	31.1	42.6	56.3	66.2	75.6	74.4	72.8	64.8	51.3	37.2	31.3		
	10%	WB	21.4	24.2	33.6	42.1	51.6	62.9	64.2	62.6	53.6	42.1	30.1	25.2		
		MCDB	25.0	27.9	39.4	52.5	62.6	73.8	72.5	70.7	59.1	47.5	33.3	28.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.3	17.0	17.4	18.7	20.4	23.1	18.3	19.0	17.6	13.9	12.7	12.5		
		MCDBR	12.8	18.2	20.2	24.5	31.2	29.8	25.7	25.9	28.2	24.0	13.1	13.1		
		MCWBR	10.2	14.1	12.9	13.3	15.2	11.7	10.8	11.7	15.5	14.2	9.4	10.9		
	5% WB	MCDBR	12.2	16.1	18.7	22.7	25.4	24.6	20.6	21.2	25.9	22.6	13.0	12.8		
		MCWBR	10.0	12.9	12.4	13.2	13.5	10.8	9.7	10.4	14.6	14.0	10.1	10.7		
(40) Clear-Sky Solar Irradiance	taub		0.274	0.308	0.351	0.387	0.406	0.399	0.419	0.403	0.361	0.341	0.308	0.276		
	taud		2.035	2.017	1.981	2.134	2.204	2.317	2.283	2.293	2.364	2.261	2.076	1.981		
	Ebn at Noon		192	232	250	262	265	268	259	256	251	221	174	154		
	Edh at Noon		21	31	42	42	42	38	38	36	29	25	20	17		
(44) All-Sky Solar Radiation	RadAvg		264	580	1013	1362	1567	1880	1671	1353	880	476	242	160		
	RadStd		28	36	79	84	168	157	118	121	111	39	30	17		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page